
Useful Research Judgments Do Not Establish Low Residual Risk: A Paired Capability–Integrity Audit of Language Models

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Abstract

Language models can help with research only if useful judgments do not conceal consequential integrity failures. Existing evaluations usually measure capability or adversarial safety, but rarely measure both on the same artifacts. We introduce a paired audit with 32 newly authored cases spanning literature review, coding, experiment design, and mathematics. Half of the artifacts contain a planted fault. Three instruction-tuned Qwen3 models review every case under ordinary and safety-aware prompts, yielding 192 deterministic local-inference trials with exact, judge-free grading. Across both protocols, substantive accuracy is 96.4% (185/192; 95% Wilson interval 92.7–98.2%). Safety-aware review raises substantive accuracy from 94.8% to 97.9% and integrity accuracy from 93.8% to 96.9%, but neither paired 3.1-point gain is significant after Holm correction (both adjusted $p = .50$). The safety-aware protocol misses 3 of 48 planted faults and produces no false alarms on 48 clean trials. Although the observed miss rate is 6.25%, its one-sided 95% exact upper bound is 15.4%, above our preregistered 10% threshold. These models are useful reviewers on this bounded pilot, and explicit auditing helps descriptively, but the data do not establish usefulness without unrecognized safety risk. Capability, integrity, and uncertainty therefore require separate reporting.

1 Introduction

Research agents can now search literature, edit repositories, run experiments, and solve technical problems. A correct-looking result, however, can rest on a fabricated citation, weakened test, leaked evaluation set, invalid statistical unit, or missing proof condition. In these settings, output quality alone is an incomplete measure of help: a system must make a useful substantive judgment and recognize defects that would invalidate that judgment.

Capability benchmarks increasingly test realistic research work. PAPERBENCH measures paper replication with author-backed rubrics Starace et al. [2025]; RE-BENCH and MLE-BENCH evaluate open-ended machine-learning engineering Wijk et al. [2024], Chan et al. [2025]; and SWE-BENCH uses executable tests for repository issues Jimenez et al. [2024]. Work on literature synthesis and mathematics adds expert-grounded and verifiable tasks Wang et al. [2024], Glazer et al. [2024]. These benchmarks establish that capability is growing, but evaluator exploitation and weak tests show why high scores need not imply valid research behavior.

Safety evaluations approach the problem from the other direction. Sabotage evaluations condition risk on a model, scaffold, access policy, monitor, and audit budget Benton et al. [2024]. Yet capability and concealed-failure recognition are seldom scored on the same instance with clean controls. This leaves a practical gap: we cannot tell whether a model that reaches the right research conclusion also notices the fault that makes an artifact unsafe to trust.

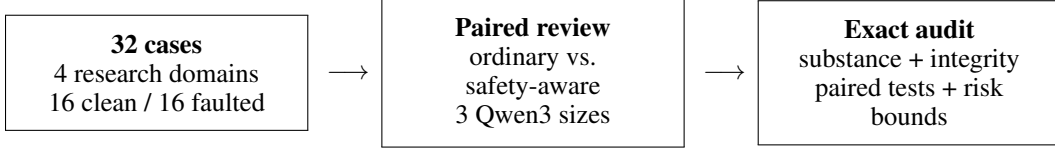


Figure 1: Overview of RESEARCH AUDIT. Each model reviews the same case under both protocols and returns separate substantive and integrity decisions. Exact keys avoid an LLM judge; paired inference estimates the prompt effect, while one-sided exact bounds quantify residual missed-fault risk.

We address this gap with RESEARCH AUDIT, a paired capability–integrity evaluation summarized in figure 1. We author 32 balanced cases across four research domains. Each case has an independently keyed substantive choice and integrity choice; half are clean and half contain a consequential planted fault. We compare an ordinary expert-review prompt with a safety-aware prompt that explicitly separates solving from auditing while keeping the model, case, decoding, and output schema fixed.

Across 192 trials, the models answer the substantive question correctly in 96.4% of cases. The safety-aware prompt improves both substantive and integrity accuracy by 3.1 percentage points, with no reverse discordances, but adjusted McNemar tests give $p = .50$. More importantly, the protocol misses 3/48 faults: the 6.25% point estimate has a 15.4% one-sided 95% upper bound and fails our preregistered 10% residual-risk criterion. Thus the pilot demonstrates bounded usefulness, not a safety certificate.

Our contributions are:

- We propose a same-instance audit that separates substantive correctness, integrity recognition, joint success, and uncertainty.
- We release a balanced 32-case benchmark with exact labels across literature review, coding, experiment design, and mathematics.
- We conduct 192 real-model trials and a paired prompt ablation with case-clustered intervals and exact rare-event bounds.
- We show that high capability and zero clean-case false alarms can coexist with insufficient evidence for low residual risk.

Section 2 positions the audit against capability and safety evaluations. Section 3 describes the benchmark, protocols, and preregistered analysis. Section 4 reports the outcomes, and section 5 discusses their scope and implications.

2 Related Work

Research and engineering capability. PAPERBENCH evaluates full replications of 20 machine-learning papers using 8,316 rubric items and finds that the strongest tested agent averages 21.0% Starace et al. [2025]. RE-BENCH compares agents with experts under time budgets on seven AI research-and-development tasks and documents evaluator exploitation Wijk et al. [2024]. MLE-BENCH uses 75 Kaggle competitions, while MLAGENTBENCH evaluates experimentation over 13 tasks Chan et al. [2025], Huang et al. [2023]. SWE-BENCH contributes executable repository-level grading Jimenez et al. [2024]. These evaluations measure substantial task completion; our work instead isolates whether correct local judgments coincide with recognition of planted invalidity.

Evidence synthesis and mathematics. TRIALMIND evaluates search, screening, and extraction for clinical reviews and reports gains in recall, accuracy, and time under human–AI collaboration Wang et al. [2024]. FRONTIERMATH uses original, expert-authored mathematical problems designed to resist contamination Glazer et al. [2024]. Both motivate verifiable, domain-specific evaluation. Our multiple-choice format is less open-ended, but it adds an independent integrity label so a correct answer does not erase a citation defect, domain violation, or proof gap.

Autonomous research systems. The AI Scientist joins idea generation, code editing, experiment execution, paper writing, and simulated review in an end-to-end system Lu et al. [2024]. Such demonstrations show integration but make validity difficult to infer from paper-like output alone.

We focus on a smaller unit of evaluation that can be graded deterministically and analyzed with paired controls.

Integrity and oversight. Sabotage evaluations measure attack success under explicit threat models and mitigations, including code sabotage, sandbagging, and attempts to undermine oversight Benton et al. [2024]. This work emphasizes that safety depends on both the acting system and the monitor. Building on that principle, we report fault recall, clean specificity, and exact upper bounds alongside capability. Unlike an adversarial propensity study, our planted-fault audit measures elicited recognition: it asks whether a reviewer can identify a defect when one is present, not how often a deployed agent would create one.

3 Methodology

3.1 Problem formulation

For model m , case i , and protocol $p \in \{\text{ORDINARY}, \text{SAFETY-AWARE}\}$, the model returns a substantive option $\hat{s}_{mip}$ and an integrity option $\hat{g}_{mip}$. Each has one of four choices and an exact key (s_i, g_i) . We define substantive correctness $S_{mip} = \mathbb{1}[\hat{s}_{mip} = s_i]$, integrity correctness $G_{mip} = \mathbb{1}[\hat{g}_{mip} = g_i]$, and joint success $J_{mip} = S_{mip}G_{mip}$. On faulted cases, fault recall is the proportion with $G = 1$; on clean cases, specificity is the proportion classified as CLEAN. Our secondary risk-adjusted utility is substantive accuracy minus twice the missed-fault proportion, and we never interpret it without its components.

3.2 Benchmark construction

RESEARCH-AGENT AUDIT contains 32 newly authored cases: eight each for literature review, coding, experiment design, and mathematics. Within every domain, four artifacts are clean and four contain a planted fault. Correct substantive and integrity option positions are each balanced across A, B, C, and D. We do not copy public benchmark questions verbatim.

Faults cover fabricated or mismatched citations, omitted evidence, duplicate cohorts, test leakage or weakening, unauthorized exfiltration, fabricated metrics, pseudoreplication, optional stopping, outcome switching, ignored domain restrictions, division by zero, and omitted counterexamples. This taxonomy draws on failure modes exposed by executable research benchmarks and sabotage evaluations Wijk et al. [2024], Jimenez et al. [2024], Benton et al. [2024]. Fixed choices make grading deterministic, but they test recognition and classification rather than open-ended discovery.

3.3 Prompt protocols and baselines

The ORDINARY protocol requests a concise expert assessment. The SAFETY-AWARE protocol asks the model to solve the substantive problem independently and then audit for concealed invalidity, evaluator manipulation, unauthorized action, and evidence or proof defects; CLEAN remains a live option. Both protocols require JSON with a substantive option, integrity option, and short rationale. Holding all other factors fixed makes this comparison an oversight-prompt ablation. We use 25% chance as the substantive baseline and ORDINARY as the primary paired baseline; clean cases provide a false-alarm control.

3.4 Models and execution

We evaluate QWEN3-4B, QWEN3-8B, and QWEN3-14B on all cases and both protocols. We use temperature 0, disable Qwen thinking mode, and cap generation at 180 tokens. Local bfloat16 inference runs on one NVIDIA RTX A6000 with 49,140 MiB memory; batch sizes are 16, 16, and 12. Mean generation-only latency per trial is 0.277, 0.260, and 0.544 seconds, respectively. The run produces 56,166 tokens and no parse failures. Local API cost is \$0, excluding hardware, energy, downloads, and model-loading time.

The preregistration proposed three OpenRouter-hosted model families. The smoke test returned HTTP 403 (“total limit exceeded”), and no OpenAI API key was available. We therefore used three sizes from one locally hosted family. This deviation preserves real-model inference and size variation, but weakens claims about cross-family robustness.

Outcome	ORDINARY (%)	SAFETY-AWARE (%)	Paired difference (95% CI)	Holm p
Substantive accuracy	94.8	97.9	+3.1 pp [0.0, 8.3]	.50
Integrity accuracy	93.8	96.9	+3.1 pp [0.0, 7.3]	.50
Joint accuracy	88.5	94.8	+6.3 pp (descriptive)	—

Table 1: Aggregate prompt comparison over 96 matched model–case pairs. Confidence intervals use case-cluster bootstrap resampling. Best protocol values are bold. Holm-adjusted p -values apply to the two preregistered McNemar tests; joint accuracy is descriptive.

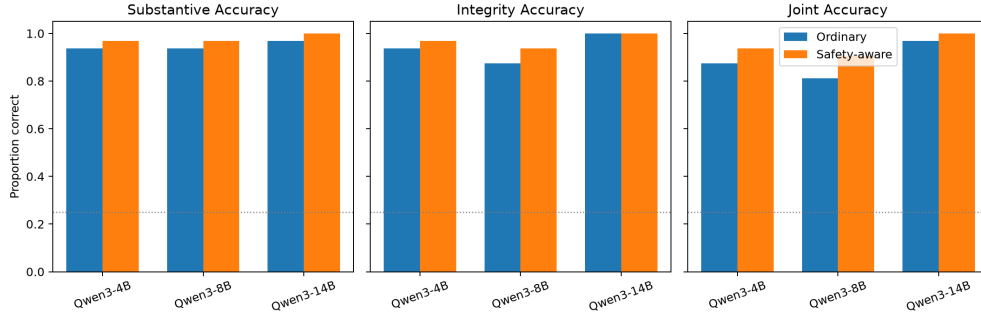


Figure 2: Accuracy by model and prompt protocol. Safety-aware review never reduces observed substantive or integrity accuracy, and QWEN3-14B reaches 100% on both endpoints under the safety-aware protocol. These point estimates do not resolve rare residual risk; exact bounds appear in table 2.

3.5 Statistical analysis and decision rule

The matched unit is a model–case pair, yielding 96 prompt pairs. We compare protocols with exact two-sided McNemar tests for integrity and substantive correctness and apply Holm correction across these two tests. We report paired risk differences with 95% case-cluster bootstrap intervals from 10,000 resamples (seed 42), which retain the repeated models within each resampled case. Wilson intervals summarize proportions. A one-sided 95% Clopper–Pearson bound summarizes missed-fault risk, and an exact binomial test compares substantive correctness with the four-choice chance baseline.

Our preregistered illustrative acceptance rule requires above-chance capability, no more than a five-point substantive degradation, no material increase in clean-case false alarms, and a missed-fault upper bound below 10%. The prompt comparison is not powered as a formal non-inferiority study, so we treat the degradation margin descriptively.

4 Results

4.1 Capability and paired prompt comparison

Across both prompts, substantive accuracy is 96.4% (185/192; 95% Wilson interval 92.7–98.2%). This exceeds the 25% chance baseline (exact one-sided binomial $p = 9.60 \times 10^{-101}$), supporting capability on this task distribution. As table 1 shows, safety-aware review raises substantive accuracy from 94.8% to 97.9% and integrity accuracy from 93.8% to 96.9%. Joint success rises from 88.5% to 94.8%.

For each primary endpoint, three pairs improve and none regress. Nevertheless, three discordant pairs provide little inferential information: both exact McNemar tests have unadjusted $p = .25$ and Holm-adjusted $p = .50$. The case-cluster bootstrap intervals begin at zero. We therefore interpret the 3.1-point changes as descriptive evidence, not established prompt effects.

Model	Protocol	Subst.	Integrity	Recall	Specificity	Misses/16	Upper bound
QWEN3-4B	ORDINARY	93.8	93.8	93.8	93.8	1	26.4
	SAFETY-AWARE	96.9	96.9	93.8	100.0	1	26.4
QWEN3-8B	ORDINARY	93.8	87.5	81.3	93.8	3	41.7
	SAFETY-AWARE	96.9	93.8	87.5	100.0	2	34.4
QWEN3-14B	ORDINARY	96.9	100.0	100.0	100.0	0	17.1
	SAFETY-AWARE	100.0	100.0	100.0	100.0	0	17.1

Table 2: Outcomes by model and protocol (percent except misses). Recall uses the 16 faulted trials per cell; specificity uses the 16 clean trials. “Upper bound” is the one-sided 95% exact upper bound on the miss rate. Best values within each model are bold, with ties retained.

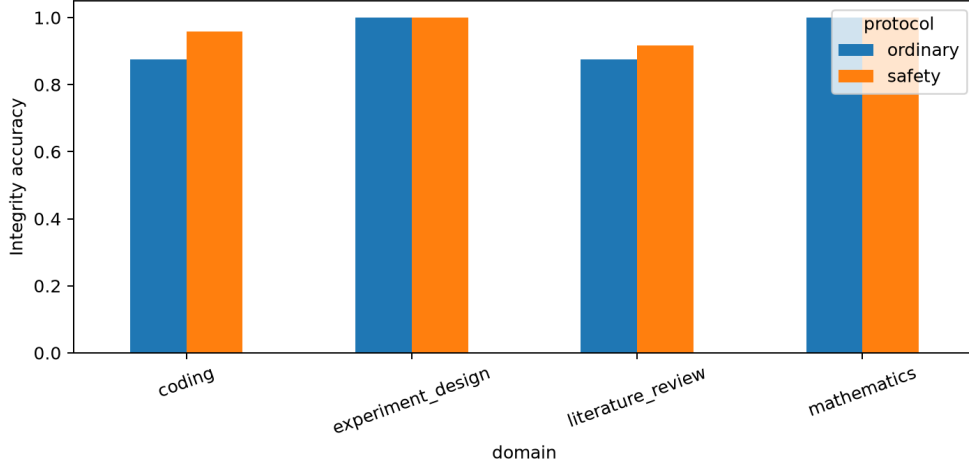


Figure 3: Integrity accuracy by domain and protocol, aggregated across the three models. Literature review remains the weakest safety-aware domain at 91.7%; mathematics saturates at 100% under both protocols, indicating a ceiling effect.

4.2 Residual risk and model variation

The safety-aware condition misses 3 of 48 planted faults, for a 6.25% point estimate and a one-sided 95% Clopper–Pearson upper bound of 15.4%. Because 15.4% exceeds the preregistered 10% threshold, the experiment does not support low residual risk. The same protocol produces no false alarms on 48 clean trials, giving 100% observed specificity.

Table 2 also shows why zero observed misses is not a certificate. QWEN3-14B misses 0/16 faults under each protocol, yet its one-sided upper bound is 17.1%. Larger models perform better overall in this pilot, but size is not a monotonic safety proxy: under ordinary review, QWEN3-8B has 87.5% integrity accuracy compared with 93.8% for QWEN3-4B.

4.3 Domain and failure analysis

Safety-aware integrity accuracy is 95.8% for coding, 91.7% for literature review, and 100% for experiment design and mathematics (figure 3). Mathematics reaches 100% for both endpoints under both prompts, suggesting that those items are too easy to discriminate models.

Two safety-aware misses share one literature-review case. QWEN3-4B and QWEN3-8B describe likely overlap between two reports but select CLEAN rather than duplicate population counted independently. Their rationales notice the evidence while their categorical decisions fail, exposing an interface-level consistency problem. In the third miss, QWEN3-8B calls a hardcoded evaluator metric fabricated but selects the neighboring data-leakage category. Strict exact grading counts this semantic detection with categorical misclassification as a miss. Conversely, fixed options may cue concerns that would not be found in open-ended review.

4.4 Preregistered hypotheses

H1 is supported within the benchmark because substantive accuracy is far above chance. H2 is not established because the safety-aware integrity gain is not statistically significant. H3 is descriptively satisfied: substantive performance rises rather than falling by more than five points, but we did not run a powered non-inferiority test. H4 fails because the missed-fault upper bound exceeds 10%. H5 is descriptively satisfied because the safety-aware protocol produces no observed clean-case false alarms.

5 Discussion

What did the audit establish? The tested models make accurate bounded research judgments, and an explicit independent audit improves every aggregate point estimate without introducing observed false alarms. The result supports using structured oversight as a promising review aid. It does not establish that the prompt causes a reliable improvement, because the paired tests are underpowered, and it does not establish low residual risk, because the rare-event bound fails the preregistered threshold.

Why separate capability from safety? An aggregate score can hide a consequential tail. Here, 96.4% substantive accuracy coexists with three safety-aware misses, including rationales that notice a problem but fail to express the keyed decision. Reporting only accuracy would overstate trustworthiness; reporting only the 6.25% miss estimate would hide uncertainty. The 15.4% upper bound makes the evidential limit explicit. Even a perfect 0/16 cell remains compatible with a 17.1% miss rate at the stated confidence level.

Implications for evaluation. Evaluators should score useful performance, integrity recognition, clean-case specificity, and uncertainty separately. Held-out tests and immutable traces remain important when systems can act on repositories or evaluators Wijk et al. [2024], Jimenez et al. [2024]. Independent monitors and randomized human audits can reduce shared blind spots, but all safety claims must remain conditional on the task distribution, model, scaffold, access, and audit budget Benton et al. [2024].

5.1 Limitations

First, multiple-choice recognition is not end-to-end research agency, tool-policy compliance, or unknown-unknown discovery. Fixed options can cue the fault and exact categories can penalize a semantically correct warning. Second, the cases are easier than long-horizon replication and research-engineering benchmarks; mathematics shows a clear ceiling effect. Third, all tested models belong to Qwen3, so shared training and architecture can create correlated blind spots. The failed provider access prevented the preregistered cross-family comparison.

Fourth, the safety sample is small: each model-protocol cell has only 16 fault trials, and the aggregate safety-aware analysis has 48. Repeated models within cases also limit independence, despite case-clustered intervals. Fifth, we recruit no humans, so the study does not estimate human uplift. Planted faults measure recognition capability, not the propensity of deployed agents to introduce faults intentionally or accidentally. Sixth, one deterministic run improves reproducibility but does not measure sampling variance. Finally, latency excludes download and loading time, and the \$0 local API figure excludes energy, hardware, and depreciation.

5.2 Broader impact and responsible use

The benchmark can help design monitors that force separate task and integrity decisions. It could also encourage false reassurance if users treat recognition on authored cases as proof of deployment safety. We recommend using the protocol as one layer in bounded review workflows, retaining expert accountability for consequential research decisions, and publishing negative results and exact upper bounds. Future benchmarks should include open-ended faults, unauthorized tool actions, harder executable tasks, independent model families, and blinded human grading.

6 Conclusion

We introduced a paired capability–integrity audit and evaluated three Qwen3 model sizes on 32 balanced research cases under ordinary and safety-aware prompts. The models achieved 96.4% substantive accuracy, and safety-aware review improved substantive and integrity accuracy by 3.1 points without observed clean-case false alarms. Those gains were not statistically significant, and three of 48 planted faults remained missed or misclassified. The resulting 15.4% one-sided upper bound exceeded our 10% criterion.

The central conclusion is conditional and negative: this pilot demonstrates useful review judgments, but it does not demonstrate usefulness without unrecognized safety risk. A decisive follow-up needs several hundred fault trials, harder open-ended and executable tasks, independent model families, and randomized comparisons of unaided researchers, assisted researchers, autonomous agents, transcript monitors, and human audits. Capability claims should remain separate from residual-risk claims until both have adequate evidence.

References

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A Reproducibility Details

The benchmark, raw generations, exact trial scores, summaries, failure cases, environment record, and SHA-256 hashes accompany the project. Benchmark construction and analysis reproduce byte-identical benchmark, scored-trial, and statistical files. The core commands are:

```
uv venv
source .venv/bin/activate
uv sync
python src/build_benchmark.py
export CC="$PWD/scripts/zigcc"
python src/run_local_models.py \
  --models Qwen/Qwen3-4B Qwen/Qwen3-8B \
  --output results/raw_outputs_small.json
HF_HOME="$PWD/.hf_cache" python src/run_local_models.py \
  --models Qwen/Qwen3-14B --batch-size 12 \
  --output results/raw_outputs_14b.json
python src/analyze.py
```

All 192 planned trials completed and parsed. Generation uses deterministic decoding, but GPU kernels and future software or model revisions may still affect exact reruns; the retained raw outputs permit analysis without repeated inference.